FARHAN HOSSAIN

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Khulna / Dhaka, Bangladesh

Software Engineer • Forward Deployment Engineering Track
Khulna University of Engineering & Technology (KUET '24)

01 // EDUCATION

Khulna University of Engineering & Technology (KUET) 2024 – Present
Bachelor of Science in Computer Science & Engineering • Batch '24 (Current 4th Semester)
Core Academic Focus: Data Structures, Algorithms, Object-Oriented Software Design, Discrete Mathematics, Database Systems, Computer Architecture.

Notre Dame College, Dhaka 2022 – 2024
Higher Secondary Certificate (HSC '24) • Science Division

02 // COMPETITIVE PROGRAMMING & ALGORITHMIC ACUMEN

CODEFORCES **163+ Solved**
Peak Rating: **1120** • Active **58-Day Streak**
Handle: muhammad_farhan_hossain

LEETCODE **156+ Solved**
Strong fluency in Dynamic Programming, Graphs (BFS/DFS), Binary Search Trees, and Two-Pointer sliding windows.

03 // TECHNICAL ARSENAL

Languages	C++20 (STL, Templates) • Python 3 • C • SQL • TypeScript • JavaScript (ES6+)
Web Automation	Playwright • JobSpy • Selenium WebDriver • BeautifulSoup4 • Curl-cffi • Asyncio • Cloudflare / Anti-Bot Bypass
Data & Analytics	Pandas • Streamlit • SQLite (WAL Mode) • NumPy • Plotly Dark • MySQL • Power BI • Relational Schema Modeling
DevOps & Cloud	GitHub Actions (CI/CD Cron) • Docker • Git / GitHub • Linux / Bash • ReportLab PDF • FastAPI (Active Learning)

04 // KEY ENGINEERING PROJECTS

LinkedIn & Multi-Platform Job Market Intelligence `Python • Streamlit • JobSpy • SQLite WAL • GitHub Actions • Plotly`

- Architected an automated multi-platform recruitment intelligence platform harvesting 2,050+ vacancies across LinkedIn, BDJobs, and Indeed with 3-tier dynamic classification.
- Implemented high-concurrency scraping pipelines, SQLite WAL mode storage, Plotly dark analytics, 3-day link verification, ReportLab PDF reports, and daily automated GitHub Actions cron automation.

বইবন্ধু AI & 21-Bookstore Live Price Comparator `FastAPI • NLP Intent • Vector Search • Python • Streamlit • Docker`

- Engineered a multi-store e-commerce intelligence SaaS comparing real-time book prices and stock availability across 21 online bookstores in Bangladesh.
- Integrated a persona-driven AI book salesman concierge powered by semantic vector search and FastAPI, delivering contextual reader recommendations from real-time scraped catalog inventories.

Amazon Books Unified Anti-Bot Scraping Pipeline `Playwright • Curl-cffi • BeautifulSoup4 • Python • MySQL • SQLite`

- Engineered a high-throughput e-commerce catalog harvester with TLS fingerprint spoofing, dynamic proxy rotation, and automated Cloudflare anti-bot mitigation.
- Extracted multi-tier metadata (ratings, reviews, ISBN, author taxonomy) with automated relational schema normalization into SQLite and MySQL databases.

Dual Bookstore Scalable Extraction Engine (Wafilife & Rokomari) `Python • Playwright • Selenium • BeautifulSoup4 • Relational SQL`

- Built automated web crawlers targeting national e-commerce book portals, navigating dynamic JavaScript pagination, session throttling, and anti-bot obstacles.
- Structured complete publisher catalogs (10,000+ items) into normalized relational schemas for cross-store price intelligence and author analytics.

05 // CAREER FOCUS & HORIZON

Actively specializing in **Forward Deployment Engineering (FDE)**, solutions architecture, and robust client-facing data extraction pipelines. Rigorously preparing foundational academic coursework in distributed systems to pursue **fully-funded international graduate research scholarships (Ph.D. / M.Sc.)** abroad.